SSH CONNECTION & IPTABLES SECURITY TEST

REPORT

Academic Cyber Security Laboratory Report

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Course: Cyber Security Practical Lab

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1. Title

Establishment and Termination of SSH Connection Using IPTables (Linux to Windows Test Environment)

2. Objective

- Establish SSH connection between Linux and Windows
- Verify authentication over network
- Demonstrate firewall control using IPTables

3. Environment Setup

- Client: Linux machine (Kali Linux)
- Target: Windows machine with SSH enabled
- Network: LAN
- Protocol: SSH

4. Methodology

Step 1: SSH Connection Establishment
- Verified IP of Windows machine
- Connected using: ssh username@windows_ip
- Successful login
Step 2: Connection Verification
- Ran commands on remote system
- Confirmed active session
Step 3: IPTables Rule Implementation
- Applied rule:
sudo iptables -A INPUT -p tcp --dport 22 -j REJECT
- SSH session was terminated

5. Results

- SSH connection established successfully
- Connection terminated after IPTables rule applied
- Subsequent connections failed

6. Findings

- SSH depends on port 22
- IPTables controls traffic at kernel level
- REJECT stops connection actively

7. Conclusion

SSH connectivity and firewall control were successfully demonstrated.

8. Recommendation

- **Always verify firewall rules**
- **Use logging during testing**
- **Avoid blocking SSH unintentionally**

STARTING THE SSH SERVER ON WINDOWS POWERSHELL

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> sshd
__PROGRAMDATA__\ssh/sshd_config: No such file or directory
PS C:\Windows\system32> Add-windowsCapability -online -name OpenSSH.Server----0.0.1.0


Path          :
Online        : True
RestartNeeded : False



PS C:\Windows\system32> start-service sshd
WARNING: Waiting for service 'OpenSSH SSH Server (sshd)' to start...
PS C:\Windows\system32> Get-Service sshd

Status   Name               DisplayName
------   ----               -----------
Running  sshd               OpenSSH SSH Server


PS C:\Windows\system32> set-service -Name sshd -StartupType Automatic
```

CHECKING THE POLICY FOR RULES

```
root@kali: ~
Session  Actions  Edit  View  Help
└─# iptables -D INPUT -p tcp --dport 22 -j REJECT

┌──(root㉿kali)-[~]
└─# iptables -L -v -n
Chain INPUT (policy ACCEPT 2383 packets, 127K bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

┌──(root㉿kali)-[~]
└─# sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination

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```

LISTENING FOR PORT 22 FROM POWERSHELL

```
Status   Name               DisplayName
------   ----               -----------
Running  sshd               OpenSSH SSH Server


PS C:\Windows\system32> set-service -Name sshd -StartupType Automatic
PS C:\Windows\system32> ncat -lvnp 22
Ncat: Version 7.99 ( https://nmap.org/ncat )
Ncat: Listening on [::]:22
Ncat: Listening on 0.0.0.0:22
PS C:\Windows\system32> ncat -lvnp 22
Ncat: Version 7.99 ( https://nmap.org/ncat )
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PS C:\Windows\system32> ncat -lvnp 22
Ncat: Version 7.99 ( https://nmap.org/ncat )
Ncat: Listening on [::]:22
Ncat: Listening on 0.0.0.0:22
PS C:\Windows\system32> ssh emma@10.0.2.15
ssh: connect to host 10.0.2.15 port 22: Connection timed out
PS C:\Windows\system32> ssh kali@10.0.2.15
ssh: connect to host 10.0.2.15 port 22: Connection timed out
PS C:\Windows\system32> ssh kali@10.0.2.15
PS C:\Windows\system32> ssh kali@10.0.2.15
ssh: connect to host 10.0.2.15 port 22: Connection timed out
PS C:\Windows\system32> ncat -lvnp 22
Ncat: Version 7.99 ( https://nmap.org/ncat )
Ncat: Listening on [::]:22
Ncat: Listening on 0.0.0.0:22
PS C:\Windows\system32>
```

```
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└─# iptables -A INPUT -p tcp --dport 22 -j ACCEPT
┌──(root㉿kali)-[~]
└─# iptables -D INPUT -p tcp --dport 22 -j ACCEPT
┌──(root㉿kali)-[~]
└─# iptables -F
┌──(root㉿kali)-[~]
└─# iptables -X
┌──(root㉿kali)-[~]
└─# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::232b:da5:c11:5a92  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:fad1:b63e:6192:eb8b  prefixlen 64  scopeid 0x0
<global>
        ether 08:00:27:3b:70:ab  txqueuelen 1000  (Ethernet)
        RX packets 65353  bytes 89556378 (85.4 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 25928  bytes 1685102 (1.6 MiB)
```